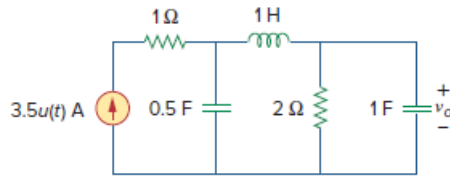


Problem

Find the voltage $v_o(t)$ in the circuit of Fig. 16.84 by means of the Laplace transform.

Figure 16.84:



Step-by-step solution

Step 1 of 7

Refer to the Figure 16.84 in the textbook.

Represent the circuit in the s-domain as shown in Figure 1.

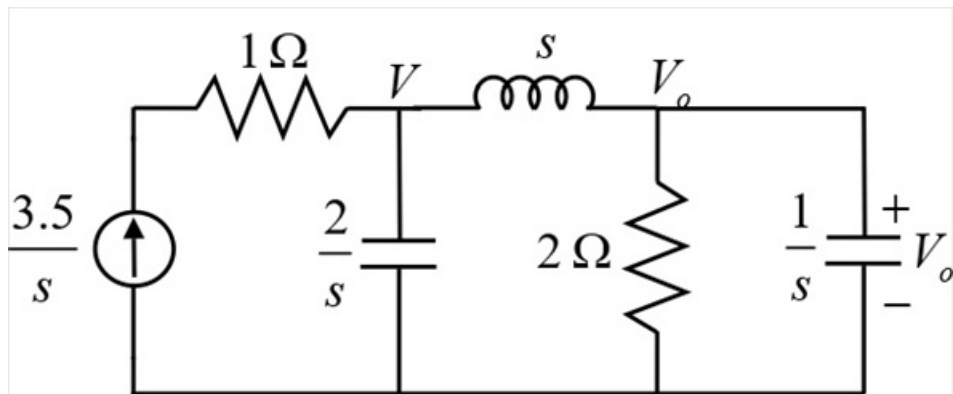


Figure 1

Step 2 of 7

Apply the nodal analysis at node V .

$$\frac{3.5}{s} = \frac{V}{\left(\frac{2}{s}\right)} + \frac{V - V_o}{s}$$

$$\frac{3.5}{s} = V \left(\frac{s}{2} + \frac{1}{s} \right) - \frac{V_o}{s}$$

$$\frac{3.5}{s} = V \left(\frac{s^2 + 2}{2s} \right) - \frac{V_o}{s}$$

$$3.5 = V(0.5s^2 + 1) - V_o \dots\dots (1)$$

Step 3 of 7

Apply the nodal analysis at node V_o .

$$\frac{V_o - V}{s} + \frac{V_o}{2} + sV_o = 0$$

$$V_o \left(\frac{1}{s} + \frac{1}{2} + s \right) - \frac{V}{s} = 0$$

$$V_o (s^2 + 0.5s + 1) - V = 0$$

$$V = V_o (s^2 + 0.5s + 1) \dots\dots (2)$$

Step 4 of 7

Substitute $V_o (s^2 + 0.5s + 1)$ for V in equation (1).

$$3.5 = V_o (s^2 + 0.5s + 1)(0.5s^2 + 1) - V_o$$

$$3.5 = V_o (0.5s^4 + s^2 + 0.25s^3 + 0.5s + 0.5s^2 + 1 - 1)$$

$$3.5 = 0.5V_o s (s^3 + 0.5s^2 + 3s + 1)$$

Rewrite the above equation.

$$V_o = \frac{7}{s(s^3 + 0.5s^2 + 3s + 1)} \dots\dots (3)$$

Step 5 of 7

Calculate the roots of the equation $(s^3 + 0.5s^2 + 3s + 1)$.

$$(s^3 + 0.5s^2 + 3s + 1) = 0$$

$$(s + 0.3395)(s^2 + 0.1605s + 2.945) = 0$$

Step 6 of 7

Rewrite the equation (3).

$$V_o = \frac{7}{s(s + 0.3395)(s^2 + 0.1605s + 2.945)}$$

Calculate the partial fraction of above equation.

$$\frac{7}{s(s + 0.3395)(s^2 + 0.1605s + 2.945)} = \frac{A}{s} + \frac{B}{s + 0.3395} + \frac{Cs + D}{s^2 + 0.1605s + 2.945}$$

Substitute 7 for A , -6.86 for B , -0.141 for C and -2.35 for D .

$$\begin{aligned} V_o &= \frac{7}{s} - \frac{6.86}{s + 0.3395} + \frac{-0.141s - 2.35}{s^2 + 0.1605s + 2.945} \\ &= \frac{7}{s} - \frac{6.86}{s + 0.3395} - \frac{0.141s}{(s + 0.0802)^2 + (1.714)^2} - \frac{2.35}{(s + 0.0802)^2 + (1.714)^2} \\ &= \frac{7}{s} - \frac{6.86}{s + 0.3395} - \frac{0.141(s + 0.0802 - 0.0802)}{(s + 0.0802)^2 + (1.714)^2} - \frac{2.35}{(s + 0.0802)^2 + (1.714)^2} \\ &= \frac{7}{s} - \frac{6.86}{s + 0.3395} - \frac{0.141(s + 0.0802)}{(s + 0.0802)^2 + (1.714)^2} - 1.32 \frac{1.714}{(s + 0.0802)^2 + (1.714)^2} \end{aligned}$$

Step 7 of 7

Take the inverse Laplace transform of the above equation.

$$v_o = \left[7 - 6.86e^{-0.33t} - 0.141e^{-0.0802t} \cos(1.714t) - 1.32e^{-0.0802t} \sin(1.714t) \right] u(t)$$

Hence, the voltage across the capacitor is

$$\boxed{\left[7 - 6.86e^{-0.33t} - 0.141e^{-0.0802t} \cos(1.714t) - 1.37e^{-0.0802t} \sin(1.714t) \right] u(t)}.$$